

# WEIYE XU

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## SUMMARY

Dual-degree M.S. student in Computer Science (University of Washington and Tsinghua University) working at the intersection of machine learning, intelligent sensing, and human-computer interaction. Three first-author papers at ACM CHI, UIST, and IMWUT, 10 patents, and research shipped to production at Ant Group; owns the full loop from hardware prototype and data collection to model design, deployment, and user study.

## EDUCATION

**University of Washington, Seattle, WA** 09/2024 – 03/2027 (expected)

M.S. in Technology Innovation (Computer Science / HCI track), Global Innovation Exchange, dual degree with Tsinghua

**Tsinghua University, Beijing, China** 09/2024 – 06/2027 (expected)

M.S. in Computer Science, Global Innovation Exchange ◇ GPA 4.0/4.0, ranked 1st in cohort ◇ Advisor: Prof. Yuanchun Shi

**Tsinghua University, Beijing, China** 09/2020 – 06/2024

B.Eng. in Computer Science and Technology ◇ GPA 3.73/4.0 ◇ Coursework: Data Structures & Algorithms, Object-Oriented Programming, Software Engineering, Artificial Intelligence, Computer Graphics, Human-Computer Interaction, Formal Languages & Automata, Digital Logic, Probability & Statistics

## HONORS & AWARDS

- Inventor on 10 granted or filed patents (9 China, 1 United States).
- National Scholarship for Graduate Students, China Ministry of Education, 2025 (top 1%).
- Outstanding Research Scholarship, Tsinghua University, 2024.

## TECHNICAL SKILLS

**Languages:** Python (expert), C/C++, JavaScript (Vue), Bash/Shell, VHDL

**Machine Learning:** PyTorch, NumPy/SciPy/Pandas; domain-adversarial learning, multimodal sensor fusion, time-series/audio modeling, LLM/RAG

**Signal Processing:** acoustic feature extraction, multi-channel audio, IMU/PPG, RF (RSSI/phase), impedance spectra

**Hardware & Systems:** PCB design (Eagle), sensor integration, multi-channel data acquisition, Arduino, 3D printing, machine knitting; Linux, Git, Docker, CI/CD, DVC

**AI Developer Tools:** Claude Code, OpenAI Codex (daily use for code generation, refactoring, and test writing)

## INDUSTRY EXPERIENCE

**Ant Group (Alipay)** 06/2026 – 09/2026

*Machine Learning Engineer Intern, Trust & Safety* Hangzhou, China

- Owned end-to-end user verification for smartwatch tap-to-pay (payment-time verification and session keep-alive): designed and ran a consented data collection protocol across 450+ sessions and built a data-cleaning pipeline that detected and repaired 100% of labeled defects.
- Trained an IMU + PPG multimodal verification model (CAME-JEPA) reaching 10% FRR at 90% TRR in a low-assurance setting, and designed the user interaction flow around it.
- Contributed the architecture survey and data preprocessing for a mixture-of-experts foundation model over heterogeneous wearable sensor streams.

**Ant Group (Alipay)** 07/2025 – 02/2026

*Machine Learning Engineer Intern, Trust & Safety* Beijing, China

- Designed and built a synchronized 16-channel microphone smart-glasses prototype with a Python data-acquisition pipeline at 96 kHz; adopted team-wide as the data collection platform for voice authentication and noise-suppression training, and reused to explore air-/bone-conduction fusion architectures.
- Developed a domain-adversarial voice anti-spoofing model reaching state-of-the-art 98% balanced accuracy; the proposed inter-channel time-delay/energy feature scheme shipped in Alipay's production voice verification stack.
- Filed 5 patents (3 granted, assigned to Ant Group and Tsinghua University).

**VITURE** 01/2025 – 04/2025

*Software Development Engineer Intern* Beijing, China

- Built a real-time YOLOv7 finger-tracking platform driven by the phone's front camera so a smartphone can act as a touch controller for AR glasses; implemented implicit calibration via progressive learning that pairs touchscreen contact events with tracked fingertip positions, delivering a working proof-of-concept endorsed by the CEO.

## RESEARCH EXPERIENCE

### **KnitID: Machine-Knitted RFID Antennas for Battery-Free Sensing and Interaction**

11/2025 – Present

*Graduate Research Assistant, WI Lab, University of Washington*

- Integrated magnet wire into the loop-over-loop structure of industrial machine knitting, shrinking conventional loop antennas by  $\sim 90\%$  while extending on-body read range by  $\sim 30\%$ ; characterized the design space (wire coating, knit structure, wale count, conductive length) with far-field simulation and empirical sweeps.
- Built a multimodal RSSI + phase sensing pipeline and a bidirectional-GRU localization model with physics-based constraints for trajectory reconstruction and real-time interaction sensing: 100% tag identification, 90%+ gesture recognition accuracy,  $<5$  cm localization error.
- Accepted as an ACM UIST 2026 poster; full paper in preparation for ACM CHI 2027.

### **AuthGlass: Voice Liveness Detection and Authentication on Smart Glasses**

07/2025 – 02/2026

*Project lead, Pervasive HCI Lab, Tsinghua University, in collaboration with Ant Group*

- Designed and fabricated two generations of a synchronized 16-channel microphone-array smart-glasses prototype (14 air-conduction and 2 bone-conduction microphones) to capture the acoustic field around the wearer's head and defend voice unlock against replay attacks.
- Collected a 42-participant genuine/replay speech dataset (AuthGlass); proposed sound-field features from inter-channel time delay and energy distribution, trained with a domain-adversarial network to reach 98% balanced accuracy.
- First-author paper accepted at ACM IMWUT 2026 (to appear); code and dataset to be open-sourced.

### **BIT: Battery-Free, IC-Less and Wireless Smart Textile Interface and Sensing System**

09/2023 – 11/2024

*Project lead, XD Lab (Simon Fraser University) and Pervasive HCI Lab (Tsinghua University)*

- Proposed a wireless-power-transfer sensing model for textiles based on multi-resonant circuits and near-field coil coupling, and the first hardware-algorithm co-design that recovers sensor state from impedance spectrum analysis.
- Fabricated a VNA-based auto-solver and several smart textile interfaces, including a glove and a shirt (Eagle PCB, soldering, auto-embroidery).
- First-author paper published at ACM CHI 2025; first inventor on a U.S. patent assigned to Florida State University.

## PROJECTS

### **CocoIndex-Code: Lightweight AST-Based Semantic Code Search (open-source contributor)**

06/2026 – Present

- Designing and implementing the text-search path (search --text, 1,000+ lines) of a Python code search CLI built on the CocoIndex incremental indexing engine (2.7k stars), targeting high-volume context retrieval for AI coding agents; iterated on query model and scope with maintainers through upstream code review.

### **PaperFriend: LLM-Powered Paper Reading Assistant**

03/2024 – 06/2024

*Course project, Department of Computer Science, Tsinghua University (advised by Prof. Jie Tang)*

- Built a retrieval-augmented generation (RAG) pipeline over arXiv with LLM-driven paper recommendation, context-aware summarization, automatic question generation for self-testing, and user background analysis.
- A 10-participant user study showed measurable gains in reading efficiency and comprehension over conventional reading, with consistently positive usability feedback.

## PUBLICATIONS (3 first-author papers at premier ACM HCI venues)

- **Weiye Xu\***, Yue Xu\*, Devin Murphy, Sen Zhang, Te-Yen Wu, Yiyue Luo. *KnitID: Machine-Knitted RFID Antennas for Battery-Free Authentication, Localization and Interaction*. ACM UIST 2026 (poster). (\*equal contribution)
- **Weiye Xu**, Zhang Jiang, Siqi Zheng, Xiyuxing Zhang, Changhao Zhang, Jian Liu, Weiqiang Wang, Yuntao Wang. *AuthGlass: Benchmarking Voice Liveness Detection and Authentication on Smart Glasses via Comprehensive Acoustic Features*. ACM IMWUT 2026 (to appear).
- **Weiye Xu**, Tony Li, Yuntao Wang, Xing-Dong Yang, Te-Yen Wu. *BIT: Battery-free, IC-less and Wireless Smart Textile Interface and Sensing System*. ACM CHI 2025.
- Devin Murphy<sup>†</sup>, Sen Zhang<sup>†</sup>, Justin Yuhang Sun<sup>†</sup>, Ghovindo Siadari, Yuxuan Miao, **Weiye Xu**, Jungwon Choi, Yiyue Luo. *Machine-Knitted Magnetoactive Textiles for Wireless Charging, Biomechanical Sensing, and Haptic Feedback*. npj Flexible Electronics (under review). (<sup>†</sup>equal contribution)
- Tianyu Yu, **Weiye Xu**, Haiqing Xu, Guanhong Liu, Chang Liu, Guanyun Wang, Haipeng Mi. *Thermotion: Design and Fabrication of Thermofluidic Composites for Animation Effects on Object Surfaces*. ACM CHI 2023.
- Tianyu Yu, Yige Fan, Zhixiang Zhang, Qingyu Hu, **Weiye Xu**, Haipeng Mi, Stefanie Mueller. *Thermaterial: Program Ambient Heat Transfer Behaviors on Objects through Fluidic Composites*. ACM CHI EA 2024.
- Zeyu Wang, Ruotong Yu, Xiangyang Wang, Jiexin Ding, Jiankai Tang, Jun Fang, Zhe He, Zhuojun Li, Tobias Röddiger, **Weiye Xu**, Xiyuxing Zhang, Huan-ang Gao, Nan Gao, Chun Yu, Yuanchun Shi, Yuntao Wang. *Computing with Smart Rings: A Systematic Literature Review*. ACM IMWUT 2025.